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Robust Multiclass Diabetic Retinopathy Classification via Hybrid Graph Attention and Convolutional Ensemble Learning

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Abstract: Diabetic retinopathy (DR), a progressive microvascular complication of diabetes, remains a leading cause of preventable blindness worldwide. Its clinical progression—from mild non-proliferative to severe proliferative stages—demands early and accurate detection to enable timely therapeutic intervention. Traditional diagnostic practices rely on the manual analysis of retinal fundus images, yet these approaches face significant limitations due to the scarcity of ophthalmic specialists and restricted access to screening, particularly in low-resource settings. This underscores the critical need for automated, scalable, and high-performing diagnostic solutions.

In response, this study introduces SCGNet, an advanced deep learning framework for automated DR classification that integrates graph-based relational reasoning, residual convolutional architectures, and a meta-learning ensemble strategy. The system is supported by a comprehensive preprocessing pipeline featuring adaptive Master-Slave artifact filtering and progressive image enhancement to optimize visual feature quality.

SCGNet was trained exclusively on the **APTOS 2019** dataset and rigorously evaluated on two external datasets—**IDRiD** and a **clinically curated private dataset**—to assess its generalizability across diverse populations and imaging conditions. On APTOS, the model achieved an accuracy of **99.45%**, precision of **99.55%**, recall of **99.12%**, F1-score of **99.12%**, Cohen’s Kappa of **99.13%**, and MCC of **99.14%**. Validation on the IDRiD dataset yielded an accuracy of **94.95%**, while the private Moroccan dataset produced an accuracy of **95.85%**, with consistently high scores across all performance metrics.

These findings highlight the model’s capacity to maintain robust diagnostic performance under domain shifts, making it a viable candidate for integration into real-world DR screening pipelines. By enabling precise, early-stage detection across

heterogeneous clinical environments, SCGNet offers promising utility in reducing the burden of diabetic vision loss and enhancing long-term patient outcomes across global healthcare systems.

Keywords— Diabetic retinopathy classification, graph neural networks, EfficientNet, ensemble learning, medical images classification, Master-Slave filtering.

1.Introduction

Diabetes mellitus is a chronic metabolic disorder characterized by the body's inability to produce sufficient insulin or utilize it effectively to regulate blood glucose levels [1]. This condition can lead to serious complications, adversely affecting vital organs such as the heart, kidneys, eyes, and nervous system. The International Diabetes Federation (IDF) has reported a significant global surge in diabetes prevalence, estimating that 537 million adults aged 20 to 79 were living with the disease in 2021—already exceeding previous forecasts for 2030 [2]. Projections now indicate that this figure will rise to 643 million by 2030 and reach 783 million by 2045, representing an increase of approximately 46% over 25 years period [2]. This rapid escalation underscores the urgent need for comprehensive public health policies, effective preventive strategies, and strengthened healthcare systems to mitigate the disease's long-term social and economic consequences. Continuous monitoring and early intervention are essential to minimize its impact and reduce the burden on individuals and healthcare infrastructures [2]. Diabetic retinopathy is a medical condition primarily caused by elevated blood sugar levels, commonly linked to diabetes. High blood sugar can damage the retina, the part of the eye responsible for processing light and sending visual signals to the brain through the optic nerve [3]. It plays a vital role in converting light into neural signals, which the brain processes to form visual perceptions. Central to retinal function is the macula, with the fovea centralis being its most crucial element for visual acuity [4]. Excess sugar is recognized to harm the eyes by blocking the small blood vessels that nourish the retina, leading to fluid leakage or bleeding [5].

Diabetic retinopathy (DR) is clinically classified into three primary stages: normal, non-proliferative diabetic retinopathy (NPDR), and proliferative diabetic retinopathy (PDR), based on the progression and severity of retinal damage [6]. NPDR, the early phase of the disease, is further divided into mild, moderate, and severe sub-stages, each characterized by the extent and nature of microvascular abnormalities. In the initial stages of NPDR, patients typically present with early retinal lesions such as microaneurysms (MAs), capillary occlusions, and dot hemorrhages (DH). As the condition advances to severe NPDR, more pronounced vascular anomalies, including small aneurysmal dilations of blood vessels, become evident and may lead to leakage onto the retinal surface [7]. The progression to PDR marks the most critical stage, where abnormal neovascularization and extensive bleeding significantly impair

vision, often resulting in irreversible vision loss. The presence, type, and distribution of retinal lesions serve as key indicators of disease severity [7].

These lesions, which manifest as small and localized features in retinal fundus images, are crucial for accurate DR grading. However, their reliable detection is challenged by factors such as low contrast, overlapping with typical anatomical structures, and variability in lesion appearance [8,9]. Microaneurysms, in particular, are among the earliest signs of DR but are frequently difficult to distinguish due to their subtle presentation and proximity to blood vessels, often leading to diagnostic uncertainty during early screening.

Accurate assessment of diabetic retinopathy (DR) severity is essential for guiding ophthalmologists in selecting appropriate therapeutic interventions to preserve vision and prevent irreversible damage [10]. Current clinical treatments focus primarily on halting or slowing disease progression, which necessitates precise and timely grading of DR. However, traditional diagnosis remains heavily reliant on manual evaluation of retinal images—a process that is both time-intensive and dependent on highly trained specialists and advanced imaging equipment [11]. These requirements pose significant barriers in underserved regions, particularly in remote or economically disadvantaged communities, where such resources are often scarce or entirely unavailable. Moreover, the quality of retinal fundus images can vary widely, and poor image resolution or contrast may contribute to the misidentification or under-detection of subtle lesions, such as microaneurysms or dot hemorrhages, further compromising diagnostic reliability [12].

Notably, around 79% of individuals affected by DR are concentrated in developing or underdeveloped nations, where the availability of basic screening tools and qualified ophthalmologists is severely limited [13]. The growing global prevalence of DR has placed additional strain on conventional manual screening methods, which are increasingly failing to meet rising diagnostic demand [14-15]. These limitations highlight the urgent need for scalable, automated, and cost-effective solutions to support early detection and monitoring in resource-constrained settings.

Deep learning has become a central paradigm in the automated analysis of diabetic retinopathy (DR), with Convolutional Neural Networks (CNNs) playing a particularly influential role in fundus image classification. One of the earliest clinically significant demonstrations was reported by Gulshan et al. [16], who employed a 26-layer CNN and achieved a sensitivity of 97.5%. Despite this strong diagnostic performance, the proposed framework addressed DR detection mainly as a binary classification problem, which limits its suitability for fine-grained severity assessment in routine clinical workflows. Subsequent systems, including IDx-DR [17], introduced image quality assessment strategies based on fractal analysis to improve screening reliability. However, their diagnostic sensitivity was still constrained when deployed in heterogeneous primary-care environments, where

image quality, acquisition protocols, and patient characteristics may vary considerably.

In response to the need for clinically meaningful assessment, recent research has increasingly focused on multiclass DR grading rather than simple referable/non-referable classification. Although multiclass grading better reflects ophthalmological practice, it introduces additional challenges, including subtle visual differences between adjacent disease stages, imbalanced class distributions, and variations in lesion morphology. For example, attention-gated Inception-v3-based approaches have shown insufficient sensitivity in identifying early DR stages [18], where accurate detection is essential for timely intervention. At the same time, several state-of-the-art models achieve competitive performance at the cost of substantial computational demand, often requiring high-memory GPU resources and complex training pipelines, thereby limiting their applicability in large-scale or resource-limited screening programs [19].

Recent deep architectures, such as ResViT FusionNet [20], RetNet-10 [21], and various hybrid CNN-Transformer frameworks [22–26], have reported accuracy values above 98%. While these results demonstrate the potential of advanced representation learning, they are frequently obtained under closed-set experimental settings, in which training and evaluation are performed on the same benchmark dataset, such as APTOS or IDRiD. Such protocols may favor the learning of dataset-dependent visual patterns, including acquisition-specific contrast, illumination, and resolution characteristics, rather than disease-related features that generalize across clinical domains. Consequently, performance may degrade when these models are tested on external datasets collected from different populations, imaging devices, and acquisition conditions [27, 28]. This limitation highlights the importance of developing DR grading systems that are explicitly designed for robustness and cross-domain generalization, rather than solely maximizing accuracy on individual benchmarks.

To overcome the restricted local receptive field of conventional CNNs, Graph Neural Networks (GNNs) have recently been explored as a means of modeling long-range dependencies and structural interactions among retinal lesions. By representing pathological regions or anatomical components as graph nodes, GNNs can potentially capture global relationships that are difficult to model using convolution alone. Nevertheless, existing CNN-GNN-based DR frameworks still face several limitations, including high computational burden, insufficiently interpretable graph construction, and limited alignment with clinically relevant disease grading. Lei et al. [29] proposed GF-CapsNet, which combines ResNet-18, graph-based modules, and Capsule Networks to represent inter-lesion dependencies. Although the method improved performance over conventional Capsule Networks on APTOS and IDRiD, it introduced additional computational complexity, and its robustness under cross-dataset evaluation was not fully demonstrated.

Similarly, Pandugula et al. [30] introduced a hybrid CNN-Graph Attention Network that improved Quadratic Weighted Kappa performance. However, the graph structure was mainly derived from feature similarity, rather than from anatomically or pathologically meaningful relationships, which may reduce clinical interpretability. Moreover, dense multi-head attention mechanisms increase inference cost and may hinder deployment in real-time screening scenarios. Deepsahith et al. [31] proposed a graph-enhanced multimodal fusion framework that integrates MobileNetV3 features with vascular graph embeddings using transformer-based cross-attention. Although the model achieved accuracy above 98% on EyePACS and APTOS, it was designed for binary classification and relied strongly on accurate vessel segmentation. This dependency may compromise robustness in low-quality fundus images, where vessel extraction is often affected by illumination artifacts, low contrast, blur, and acquisition noise.

Overall, existing studies demonstrate substantial progress in deep learning-based DR diagnosis, but several open challenges remain. In particular, many approaches prioritize benchmark-specific performance while giving limited attention to external validation, computational efficiency, graph interpretability, and clinically consistent multiclass grading. These gaps motivate the development of lightweight and generalizable DR grading models capable of capturing both local lesion features and global retinal relationships across heterogeneous imaging conditions.

To address the aforementioned limitations in diabetic retinopathy (DR) classification, we propose Stacked Convolutional and Graph Attention Network (**SCGNet**). This novel hybrid framework combines dual-path probabilistic modeling with ensemble stacking to enhance diagnostic accuracy and generalizability. The architecture comprises two parallel and complementary branches, each designed to capture distinct feature representations and independently estimate class probabilities. This dual-path structure enables the model to extract richer and more diverse information from retinal images.

This study presents several novel contributions that collectively advance the state of automated diabetic retinopathy (DR) detection:

1. **Multiclass Classification Capability:** SCGNet is designed to perform fine-grained multiclass classification of DR, surpassing the binary classification frameworks commonly used in prior literature. This enables more precise staging of disease severity, aligning more closely with clinical diagnostic requirements.
2. **Cross-Dataset Generalization Without Domain Adaptation:** Unlike many existing approaches that require domain-specific tuning, SCGNet demonstrates strong generalization by being trained exclusively on the APTOS 2019 dataset and evaluated—without retraining—on two independent and demographically distinct datasets. This addresses a critical limitation in current AI models related to poor adaptability under domain shift.

3. **Consistent Performance and Overfitting Mitigation:** The framework achieves high and stable performance across both internal and external evaluations. This robustness reflects a reduced tendency toward overfitting and confirms the reliability of the model under varied conditions.
4. **Resilience to Image Quality Variations:** SCGNet integrates an advanced preprocessing pipeline, incorporating adaptive artifact filtering and progressive enhancement, which ensures strong diagnostic accuracy even when processing low-quality or heterogeneous fundus images—a common challenge in real-world screenings.
5. **Validation on Multi-Ethnic, Real-World Clinical Data:** The model is explicitly validated on a diverse dataset composed of retinal images from Indian and Moroccan populations. This demographic heterogeneity allows for a thorough evaluation of diagnostic consistency across variations in anatomical features and pigmentation levels, demonstrating SCGNet's suitability for equitable deployment across global healthcare systems.
6. **Parameter-efficient ensemble learning:**

Unlike computationally intensive end-to-end training strategies used in GF-CapsNet [29] and transformer-based fusion models [31], SCGNet adopts a parameter-efficient stacking framework. In this design, high-capacity pre-trained feature extractors, namely EfficientNet-B3a and ResNet50, are retained as frozen backbone networks, while optimization is restricted to a small set of downstream classification and fusion components. Consequently, the proposed model requires fewer than 0.11 million trainable parameters. This substantially reduces the optimization burden compared with recent hybrid deep learning architectures, with an approximate two-order-of-magnitude decrease in trainable parameters. Such a compact design enhances computational efficiency, lowers memory requirements, and supports deployment in resource-constrained clinical environments, while preserving competitive diagnostic performance.

The remainder of this paper is structured as follows: Section 2 describes the materials and methods, including details of the retinal image datasets, preprocessing procedures, and the proposed model architecture with ensemble stacking. Section 3 outlines our experimental design, which includes benchmarking against state-of-the-art models, ablation studies to validate each component's contribution, and a thorough performance analysis using both quantitative metrics and qualitative visualizations. Finally, Section 4 presents our conclusions, discussing the clinical relevance of SCGNet in DR screening and outlining future research directions, particularly in scalable, multimodal, and graph-based medical image analysis.

2. Materials and methods

This section describes the methodological design of the proposed framework for automated diabetic retinopathy (DR) severity grading. The framework is organized

into two complementary components: an adaptive retinal image preprocessing module and a hybrid stacked learning architecture.

The first component, presented in Section 2.2, consists of a multi-stage preprocessing pipeline designed to improve image consistency and enhance lesion-relevant retinal structures. Initially, raw RGB fundus images are decomposed into their individual color channels to enable channel-specific spectral processing. Each channel is then normalized using min-max intensity scaling to reduce illumination variability and standardize pixel-value distributions across images acquired under different imaging conditions. Subsequently, the proposed Channel Adaptive Contrast and Noise Suppression (CACNS) algorithm is applied independently to each channel. CACNS is formulated as an adaptive recursive filtering mechanism that progressively attenuates high-frequency noise while preserving fine pathological patterns, particularly microaneurysms, hemorrhages, and lesion boundaries.

Within each CACNS processing block, contrast enhancement is first performed using Contrast Limited Adaptive Histogram Equalization. This step improves local visibility while limiting excessive amplification of background noise. Residual noise is then reduced through bilateral filtering, which preserves edge information by jointly considering spatial proximity and intensity similarity. Finally, a residual enhancement operation reintroduces selected high-frequency structural information to maintain the morphology of small retinal lesions and prevent over-smoothing. After channel-wise enhancement, the processed channels are recombined, transformed into grayscale images, and resized to a fixed spatial resolution of 256×256 pixels. This standardization provides a consistent input representation for the subsequent learning stage.

The second component, detailed in Section 2.3, introduces SCGNet, a Stacked Convolutional and Graph Attention Network designed to integrate complementary visual and relational representations. SCGNet follows a dual-branch architecture. In the first branch, EfficientNet-B3 is used as a convolutional feature extractor to obtain discriminative retinal descriptors. These features are subsequently reduced using Principal Component Analysis to decrease dimensionality and computational cost. The compact feature space is then used for global clustering through K-means, followed by k-nearest-neighbor graph construction. A Graph Attention Network is applied to the resulting graph in order to model inter-sample relationships through adaptive attention-based message passing.

The second branch employs ResNet-50 to capture hierarchical visual representations from the preprocessed fundus images. Similar to the first branch, the extracted feature vectors are compressed using Principal Component Analysis before classification, thereby improving computational efficiency and reducing redundancy. The class-probability outputs generated by both branches are then combined using an XGBoost meta-classifier within a stacking ensemble strategy. This fusion mechanism enables the model to exploit both graph-based relational dependencies and convolutional image-level features.

To mitigate the effect of class imbalance, which is a common challenge in DR grading datasets, the training procedure incorporates dynamic loss reweighting based on inverse class frequency. This strategy increases the contribution of underrepresented severity classes during optimization. In addition, photometric data augmentation and dropout regularization are employed to improve robustness and reduce overfitting. The section also presents the adopted multi-dataset evaluation strategy, including experiments on APTOS 2019, IDRiD, and a Moroccan clinical dataset. Furthermore, computational profiling, hyperparameter settings, and data leakage prevention procedures are reported to ensure transparency, reproducibility, and methodological validity.

Overall, the proposed methodology is designed to produce standardized lesion-preserving retinal representations while integrating convolutional and graph-based learning in a parameter-efficient stacked framework. This design aims to improve multiclass DR grading performance, enhance robustness across heterogeneous datasets, and support practical deployment in resource-constrained clinical screening environments.

2.1. Dataset Description and Justification for Multi-Source Selection

This study leverages three distinct retinal fundus image datasets, each selected for its unique geographic, demographic, and technical characteristics. This multi-source approach is instrumental in building a diabetic retinopathy (DR) classification framework capable of robust performance across diverse clinical settings and patient populations.

The APTOS 2019 Blindness Detection dataset (Table 1) serves as the primary training source and comprises 3,662 retinal images collected by the Aravind Eye Hospital in rural India. A hallmark of this dataset is the considerable heterogeneity in image quality, which arises from the use of various fundus cameras and diverse acquisition conditions encountered in field clinics. This variability, rather than being a drawback, serves as a critical strength. It challenges the model to extract stable and generalizable pathological features despite noise, poor illumination, or low contrast, closely reflecting the unpredictability of real-world screening scenarios. All images were expertly labeled by certified ophthalmologists according to the International Clinical Diabetic Retinopathy Disease Severity Scale (ICDR-DSS), with annotations spanning five ordinal categories: 0 (No DR), 1 (Mild NPDR), 2 (Moderate NPDR), 3 (Severe NPDR), and 4 (Proliferative DR). These high-quality annotations provide a consistent standard for training and evaluating multiclass classification models.

To complement APTOS, the Indian Diabetic Retinopathy Image Dataset (IDRiD) (Table 2) was integrated into the validation pipeline. Comprising 516 high-resolution images, IDRiD was acquired under controlled clinical conditions using uniform imaging protocols. Unlike APTOS, this dataset features minimal variability in illumination and focus, making it particularly well-suited for algorithmic

benchmarking and performance evaluation under ideal imaging conditions. Its consistent quality allows for precise assessments of a model's diagnostic capabilities without interference from acquisition artifacts.

The third dataset (Table 3) is a private clinical collection assembled under the supervision of Dr. Bahia Ouazzani Chahdi [32] at Cheikh Zaid Hospital in Rabat, Morocco. This dataset includes 313 images from a Moroccan patient cohort, obtained through a standardized diabetic retinopathy screening protocol. Retinal photographs were captured using Topcon and Eidon fundus cameras, following a full ophthalmological workup—including visual acuity testing, slit-lamp anterior segment evaluation, and dilated fundus examination. Grading was performed under direct medical supervision in accordance with the ICDR-DSS, and the presence of diabetic macular edema (DME) was also documented. This dataset introduces significant geographic and phenotypic diversity, including varied pigmentation patterns and anatomical differences, which are essential for evaluating model generalizability across ethnic groups. The institutional review board granted ethical approval for data collection, and all participants provided written informed consent.

By incorporating datasets that vary in population characteristics, image acquisition protocols, and overall image quality, this study adopts a training and evaluation strategy that more closely approximates the diversity encountered in real-world clinical screening environments. Such heterogeneity is essential for assessing the generalizability of automated diabetic retinopathy grading systems beyond dataset-specific conditions.

A patient-level data partitioning strategy was applied across all three datasets. Accordingly, retinal images belonging to the same individual were assigned exclusively to either the training subset or the testing subset, ensuring that no patient contributed images to both partitions. This procedure minimizes the risk of data leakage and prevents an overestimation of model performance caused by patient-specific visual similarities.

Through this unified experimental design, the proposed framework is evaluated not only in terms of classification accuracy under controlled benchmark settings, but also with respect to its robustness, external consistency, and potential applicability across diverse clinical environments. This approach provides a more rigorous assessment of the model's capacity to operate reliably under heterogeneous real-world conditions.

Table 1: APTOS 2019 Training Dataset Specifications

Parameter	Details
Source	APTOS Blindness Detection Challenge
Total Images	3,662
Image Resolution	Varies (typically 1736×1824px)

Class Distribution	0:Normal	1805	(49.3%)
	1:Mild_DR	370	(10.1%)
	2:Moderate_DR	999	(27.3%)
	3:Severe_DR	193	(5.3%)
	4:Profilative_DR	295	(8.0%)

Table 2: IDRiD Validation Dataset Specifications:

Parameter	Details		
Source	Indian Diabetic Retinopathy Image Dataset		
Total Images	516		
Image Resolution	4288×2848px		
Class Distribution	0:Normal	331	(64.1%)
	1:Mild_DR	73	(14.1%)
	2:Moderate_DR	57	(11.0%)
	3:Severe_DR	35	(6.8%)
	4:Profilative_DR	20	(3.9%)

Table 3: Private Dataset Specifications:

Parameter	Details		
Source	Private National Dataset		
Total Images	313		
Image Resolution	1100×828px		
Class Distribution	0:Normal	83	(26.5%)
	1:Mild_DR	41	(13.1%)
	2:Moderate_DR	84	(26.8%)
	3:Severe_DR	83	(26.5%)
	4:Profilative_DR	22	(7.0%)

Ethic aspects:

The Cheikh Zaid Foundation supported this study. Accordingly, it was approved by the Local Ethics Committee of Cheikh Zaid Hospital, Rabat, Morocco, Project: CEFCZ/PR/03/06/24.

2.2. Image Preprocessing Pipeline for Retinal Fundus Images

The performance of automated diabetic retinopathy (DR) grading systems is strongly influenced by the visual quality and consistency of retinal fundus images. In practical screening environments, fundus images are often affected by acquisition-related degradations, such as non-uniform illumination, reduced contrast, sensor noise, and fine-scale artifacts. These factors may conceal clinically important retinal abnormalities, including microaneurysms, hard exudates, hemorrhagic regions, and subtle vascular alterations, thereby complicating reliable feature extraction and classification.

To mitigate these limitations while maintaining the integrity of diagnostically meaningful retinal structures, this study introduces a multi-stage adaptive preprocessing pipeline. The proposed procedure is designed to enhance image quality, suppress irrelevant noise, and preserve lesion morphology before model training and inference. As illustrated in Fig. 1, the workflow progressively transforms raw RGB fundus images into standardized grayscale representations with a spatial resolution of 256×256 pixels, which are subsequently used as input for the proposed deep learning-based DR classification framework.

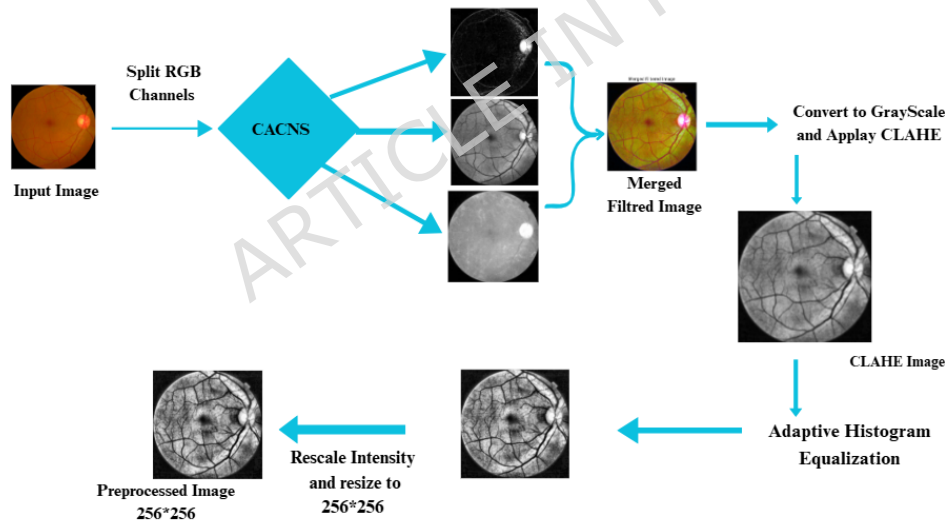


Figure 1. Overview of the proposed fundus image preprocessing framework. The input RGB fundus image is first decomposed into its individual color channels, each of which is processed using the CACNS algorithm (the detailed architecture is presented in Figure 2). The enhanced channels are then recombined to reconstruct the image. Subsequently, the reconstructed image is converted to grayscale, further enhanced through Contrast Limited Adaptive Histogram Equalization (CLAHE), and finally resized to 256×256 pixels to produce a standardized preprocessed output.

The proposed preprocessing strategy differs from conventional global enhancement techniques by exploiting the channel-dependent spectral information inherent in color fundus imaging. As shown in the initial stage of Fig. 1, each RGB retinal image is decomposed into its red, green, and blue channels prior to enhancement. This separation is justified by the fact that each spectral component captures different anatomical and textural information from the retina.

The red channel is generally associated with deeper light penetration and may provide useful information related to choroidal and deeper vascular structures. The green channel typically offers higher contrast for superficial retinal vessels, the nerve fiber layer, and several clinically relevant lesions, making it particularly informative for DR analysis. In contrast, the blue channel is more sensitive to light scattering and noise; however, it may still retain complementary surface-level texture information that contributes to retinal characterization. By processing each channel separately, the proposed framework enables adaptive enhancement according to the specific information content and degradation profile of each component. This strategy reduces undesirable cross-channel interactions and limits the color and contrast distortions that may arise when uniform global operations are applied directly to the complete RGB image.

Following spectral decomposition, each channel is independently normalized using min-max intensity scaling. This operation standardizes the dynamic range of pixel intensities and reduces variability caused by differences in illumination, camera settings, and acquisition conditions. By rescaling pixel values to the interval $[0,1]$, normalization provides a consistent numerical representation for the subsequent adaptive filtering and enhancement stages. For each spectral component, the normalized intensity is computed as:

$$I_{\text{norm}} = \frac{I_c - I_{\min}}{I_{\max} - I_{\min}}$$

where I_c denotes the intensity value of a pixel in the considered channel, while $I_{\min}$ and $I_{\max}$ represent the minimum and maximum intensity values within that channel, respectively. This normalization ensures that all channel-specific inputs contribute within a comparable intensity range before being processed by the proposed adaptive enhancement pipeline.

Noise attenuation is performed using the proposed adaptive master-slave recursive filtering scheme, whose architecture is presented in Fig. 2. In contrast to conventional fixed-parameter filtering methods, the proposed approach employs an iterative feedback mechanism that adaptively regulates the trade-off between noise suppression and anatomical structure preservation. This design is particularly important in diabetic retinopathy analysis, where excessive smoothing may eliminate small but clinically relevant lesions, such as microaneurysms and fine vascular abnormalities.

At the i -th iteration, an error term e_i is first computed by measuring the difference between the normalized input image and the filtered output obtained from the previous iteration, denoted as y_{prev} . This error feedback guides the filter toward progressively reducing residual noise while maintaining consistency with the original retinal signal. In parallel, spatial regularization is introduced through a Laplacian operator applied to the previous output:

$$L_i = \nabla^2 y_{\text{prev}}$$

This term promotes local structural continuity and contributes to the preservation of edge-related information. The recursive update is then formulated as:

$$y_i = 2\sigma\alpha e_i - \alpha^2 y_{\text{prev}} + 0.5L_i$$

where σ denotes the smoothing parameter and α represents the adaptation coefficient. Unlike static filtering strategies, these parameters are not predefined globally. Instead, they are estimated dynamically from local noise statistics measured within homogeneous retinal regions. This adaptive parameterization allows the filter to respond to spatial variations in noise distribution and image texture.

As illustrated in Fig. 2, the recursive procedure generally reaches convergence within 15–30 iterations. This enables effective suppression of high-frequency noise while reducing the risk of over-smoothing diagnostically significant structures. Consequently, the proposed filtering mechanism preserves the morphology of microlesions, vessel contours, and fine capillary patterns, which are essential for reliable DR grading [33–35].

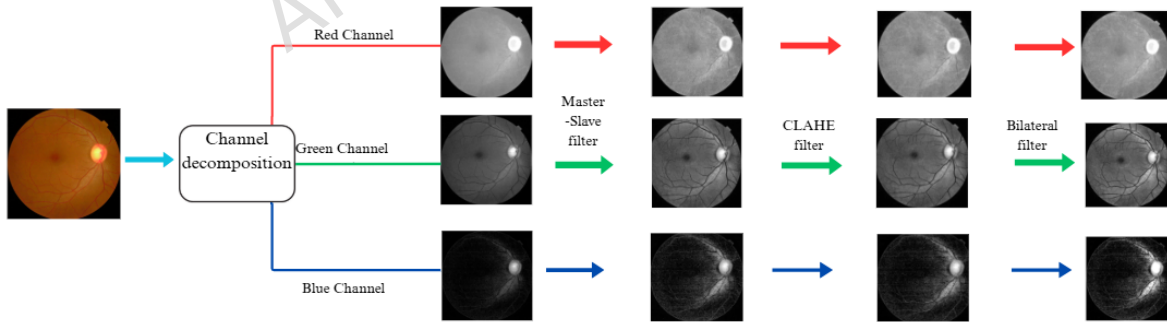


Figure 2. Internal architecture of the CACNS (Channel Adaptive Contrast and Noise Suppression) enhancement block. The input spectral channel first undergoes master-slave adaptive filtering to suppress structural noise. The resulting signal is then processed using Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance contrast, particularly in low-illumination regions. Finally, bilateral filtering is applied to remove residual

high-frequency noise while preserving the sharp boundaries of micro-lesions and vessel walls.

To address the low and spatially non-uniform contrast commonly observed in retinal fundus images, the denoised spectral channels are subsequently processed using Contrast-Limited Adaptive Histogram Equalization (CLAHE) [37, 38]. Unlike global histogram equalization, which applies a uniform intensity transformation to the entire image, CLAHE operates locally by partitioning the image into contextual regions and redistributing pixel intensities within each region. The contrast-limiting mechanism prevents excessive amplification of background noise and avoids the generation of artificial artifacts, thereby improving the visibility of subtle retinal lesions and vascular structures.

Following contrast enhancement, a bilateral filtering operation is applied to further suppress residual noise while preserving important anatomical boundaries [39–41]. This edge-preserving filter combines spatial proximity and intensity similarity, allowing pixels with similar gray-level values and close spatial locations to contribute more strongly to the filtering process. As a result, noise is attenuated in relatively homogeneous regions, while sharp transitions corresponding to vessel borders, lesion margins, and retinal structures are retained. The bilateral filtering operation is defined as follows:

$$\text{by: } I_{\text{bilat}}(x) = \frac{1}{W_p} \sum_{x_i \in \Omega} G_{\sigma_s}(\|x_i - x\|) G_{\sigma_r}(|I(x_i) - I(x)|) I(x_i)$$

where G_{σ_s} and G_{σ_r} are Gaussian kernels controlling spatial proximity and intensity similarity, respectively.

To minimize the risk of suppressing fine pathological details during the smoothing process, a residual enhancement stage is introduced after bilateral filtering. This step is designed to recover high-frequency information that may correspond to clinically relevant microstructures, such as lesion margins, small vascular irregularities, and subtle textural variations.

For this purpose, a residual difference map D is computed by measuring the absolute pixel-wise difference between the original normalized image and the bilaterally filtered output:

$$D = |I_{\text{original}} - I_{\text{bilat}}|$$

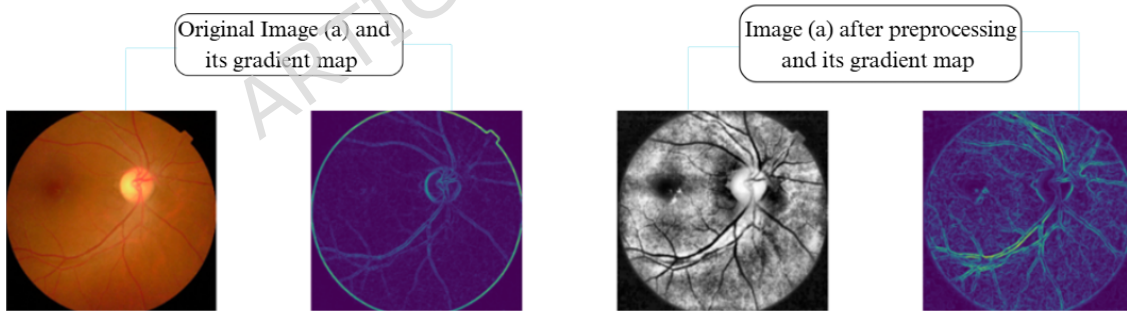
where I_{original} denotes the normalized input image before bilateral filtering, and I_{bilat} represents the corresponding edge-preserving filtered image. The resulting difference map highlights fine-scale structural components that may have been attenuated during noise reduction. After normalization, this residual information is reintegrated into the filtered image in order to selectively reinforce diagnostically

important boundaries and preserve the morphology of small retinal lesions. The final enhanced image is obtained as follows:

$$I_{\text{enhanced}} = I_{\text{bilat}}(1 + D_{\text{norm}}) \text{ where } D_{\text{norm}} = \frac{D}{225}$$

In the final preprocessing stage, the enhanced spectral components are merged to reconstruct the processed color fundus image. To reduce spectral redundancy and encourage the classification model to focus primarily on anatomical and lesion-related morphology rather than color-dependent variations, the reconstructed image is subsequently transformed into a grayscale representation. An additional adaptive contrast enhancement step is then applied to further improve the visibility of clinically relevant structures. Finally, the image intensities are rescaled to the standard 8-bit range (0-255), and all images are resized to a fixed spatial resolution of 256×256 pixels to ensure input consistency for the deep learning model.

The qualitative impact of the proposed preprocessing pipeline is illustrated in Fig. 3, where representative raw and processed fundus images are compared across different DR severity stages. The visual results indicate that the proposed approach improves local contrast, enhances retinal structure definition, and increases the apparent signal-to-noise ratio. In particular, subtle pathological patterns, including micro-lesions, hemorrhagic regions, and vascular tortuosity, become more distinguishable after preprocessing. These enhancements are important for downstream DR grading, as they provide standardized and lesion-preserving image representations that may facilitate more reliable feature extraction and improve classification robustness.



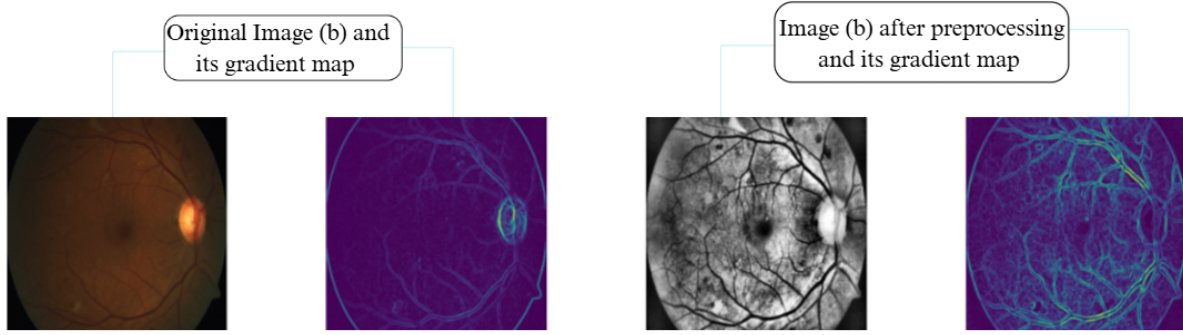


Figure 3. Comparison between original and preprocessed fundus images for diabetic retinopathy (DR) classes 0 and 1. Panels (a) and (b) correspond to the original fundus images of classes 0 and 1, respectively, along with their associated gradient maps, the corresponding preprocessed images, and the gradient maps of the preprocessed outputs. The gradient representations highlight edge information and local intensity variations. As observed, the preprocessed images generate sharper and more clearly defined gradient patterns, thereby improving the visibility of retinal structures across both classes.

2.3. Deep Learning Model Architectures for Retinal Image Analysis

Convolutional Neural Networks (CNNs) have long been the foundation of deep learning approaches in medical image analysis due to their capacity to hierarchically extract spatial features using localized convolutional filters [42]. Architectures such as ResNet [43] have consistently achieved state-of-the-art performance in various image classification tasks, including diabetic retinopathy detection. However, a core limitation of standard CNNs lies in their constrained receptive fields and their inability to explicitly model relational dependencies across image regions or between samples. This makes them less effective for capturing non-local contextual information, which can be critical in identifying subtle or spatially dispersed pathological patterns.

To address this gap, Graph Neural Networks (GNNs) have gained attention for their ability to represent and process structured data by modeling entities as nodes and their interactions as edges [44]. Among these, Graph Attention Networks (GATs) [45] introduce an attention-based mechanism that adaptively learns the relative importance of neighboring nodes. This enables more expressive and flexible modeling of relationships, allowing the network to focus on the most relevant connections during message passing. Despite their strength in relational reasoning, GNNs alone are not ideally suited for learning rich visual hierarchies from raw image data.

To leverage the complementary strengths of these paradigms, we introduce SCGNet (Stacked Convolutional and Graph Attention Network), a novel dual-branch hybrid architecture (illustrated in Figure 4). SCGNet integrates convolutional feature extraction with graph-based relational learning in a unified framework. The CNN branch captures multi-scale spatial and textural features from the input retinal images. In contrast, the GAT branch operates on graph-structured representations to model dependencies among extracted regions or lesions. These dual branches are fused via an adaptive stacking ensemble strategy, enabling the model to combine visual semantics with topological awareness to improve classification performance. This hybrid architecture is specifically designed to address the complexities of diabetic retinopathy grading, in which both local details and inter-lesion relationships are diagnostically relevant.

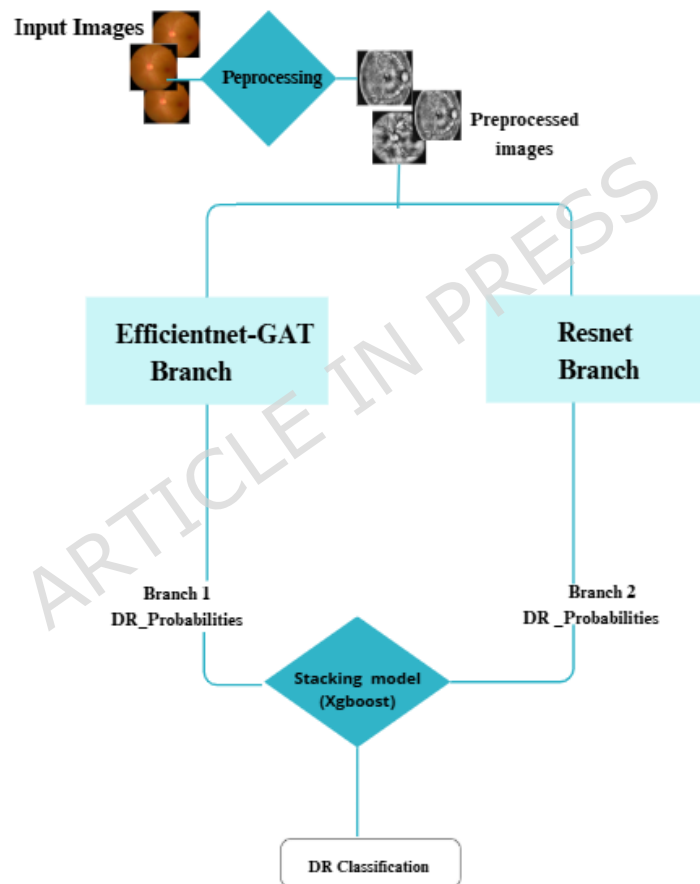


Figure 4. Global Architecture Integrating EfficientNet-GAT, ResNet, and Stacking Ensemble: SCGNet.

2.3.1.Branch 1: Graph-Based Relational Learning and Feature Modeling

The first branch of the proposed SCGNet architecture (illustrated in Figure 5) is designed to capture both fine-grained visual representations and relational dependencies among samples. This dual capability allows the model to leverage not

only the discriminative information within each retinal fundus image but also the dataset's topological structure. The processing pipeline of this branch includes multiple stages: feature extraction, dimensionality reduction, global clustering, local graph construction, graph-based reasoning, and probability computation.

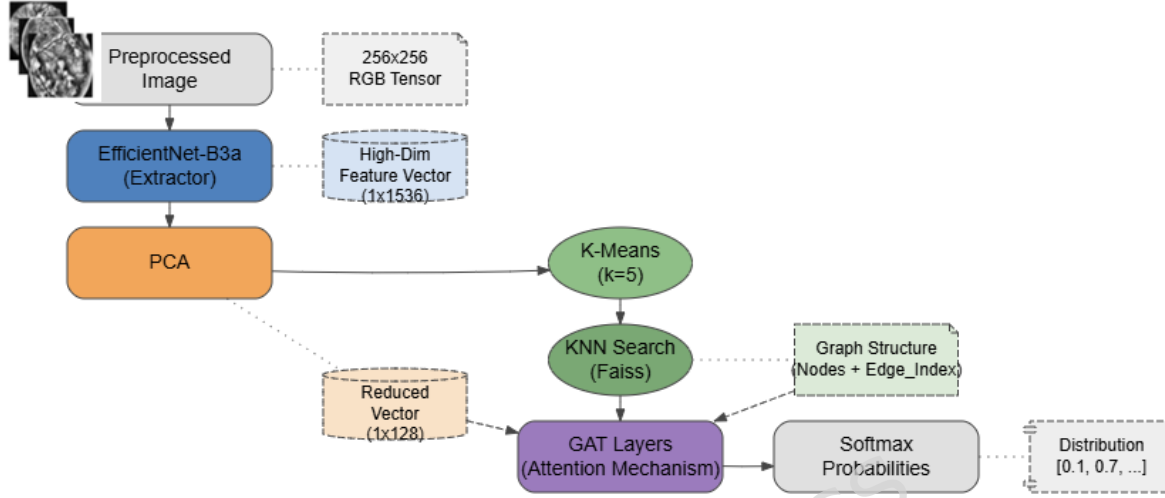


Figure 5: Illustration of the EfficientNet-GAT Branch for Feature Extraction, Graph Construction, and Probabilistic Prediction.

Each input image is first passed through EfficientNet-B3 [46], a convolutional neural network architecture known for balancing accuracy and computational cost through compound scaling of network depth, width, and resolution. For each image i , EfficientNet-B3 outputs a high-dimensional feature embedding $h_i \in \mathbb{R}^D$, where $D = 1536$. These embeddings encode multilevel semantic information:

- Low-level features: such as edges, textures, and fine-grained patterns
- Mid-level features: including anatomical motifs and small pathological structures (e.g., microaneurysms, exudates)
- High-level features: capturing global configurations, disease progression patterns, or complex retinal abnormalities

This hierarchical feature representation enables robust discrimination across varying stages of diabetic retinopathy. The selection of EfficientNet-B3 over other architectures is motivated by its excellent performance-to-efficiency trade-off, making it well-suited for hybrid frameworks that incorporate graph-based reasoning downstream.

To reduce the computational burden and avoid the curse of dimensionality in subsequent graph construction, a dimensionality reduction step is applied.

The embeddings generated by EfficientNet-B3 are high-dimensional and may contain redundant or non-informative components. To address this, Principal

Component Analysis (PCA) [47–49] is employed to project each vector $h_i \in \mathbb{R}^{1536}$ into a more compact representation $h_i \in \mathbb{R}^d$, with $d = 128$. PCA retains the most significant variance components while discarding noise and irrelevant dimensions, thereby preserving both fine-grained and high-level semantic features necessary for accurate classification. This reduction improves both computational efficiency and robustness in downstream graph-based modeling.

To characterize the coarse-level organization of the feature space and promote semantically consistent graph construction, the PCA-reduced embeddings were partitioned using K-means clustering [50]. This unsupervised step groups fundus images exhibiting similar visual patterns and latent semantic characteristics into M clusters, thereby introducing a global structural prior before the construction of local graph neighborhoods.

In the present study, the number of clusters was set to $M = 5$. This setting was selected based on both clinical relevance and empirical validation. Clinically, the choice of five clusters is consistent with the standard five-grade diabetic retinopathy severity scale, ranging from no apparent DR to proliferative DR. This correspondence enables the graph structure to better reflect the progressive nature of the disease. Empirically, the cluster number was examined through a sensitivity analysis over the range $M \in [2, 12]$, as reported in Fig. 6. Among the tested configurations, $M = 5$ achieved the highest Silhouette Score (-0.0105), indicating the most favorable trade-off between intra-cluster compactness and inter-cluster separation within the PCA-reduced representation space.

It is important to note that the clustering procedure was not used as a direct classification strategy. Instead, it served as an intermediate mechanism to guide the construction of local graph neighborhoods. By restricting k-nearest-neighbor connections to samples belonging to more homogeneous feature regions, the proposed strategy reduces the formation of noisy or semantically inconsistent edges between dissimilar retinal images. Consequently, the resulting graph connectivity becomes more coherent and better suited for subsequent graph attention-based relational learning.

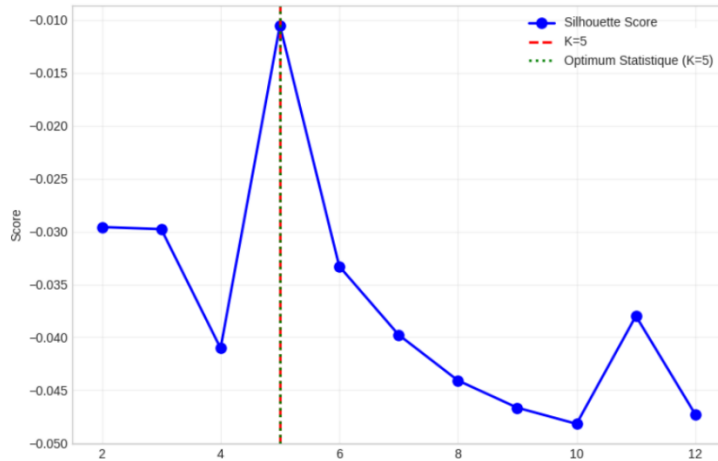


Figure 6. Empirical justification for the selection of the clustering parameter K . The plot illustrates the variation in Silhouette Score (blue curve) across different values of K , ranging from 2 to 10. The vertical dashed line marks the selected value, $K = 5$, which corresponds to the highest Silhouette Score (0.42). This choice is also clinically meaningful, as it is consistent with the five standard diagnostic stages of diabetic retinopathy (Grades 0–4).

Within each K -means cluster (Figure 7.a), a k -nearest neighbor (kNN) graph [51] is constructed to model local relationships between images (Figure 7.b). Each node in the graph corresponds to an image represented by its PCA-reduced feature vector $h_i \in \mathbb{R}^{128}$, and edges are formed by connecting each node to its k -most similar neighbors, as measured by Euclidean distance in the embedding space.

To efficiently construct these graphs in large-scale datasets, the framework employs FAISS (Facebook AI Similarity Search) [52], a high-performance library optimized for dense vector similarity search. FAISS enables exact nearest neighbor retrieval using advanced indexing structures, significantly accelerating graph construction without sacrificing accuracy. This approach ensures that the resulting graphs preserve meaningful visual relationships while maintaining scalability.

The resulting kNN graphs serve as input to the Graph Attention Network component of SCGNet, enabling localized message passing and relational reasoning within structurally coherent subspaces. This multi-stage process balances computational efficiency and model expressiveness, forming a strong foundation for robust diabetic retinopathy classification.

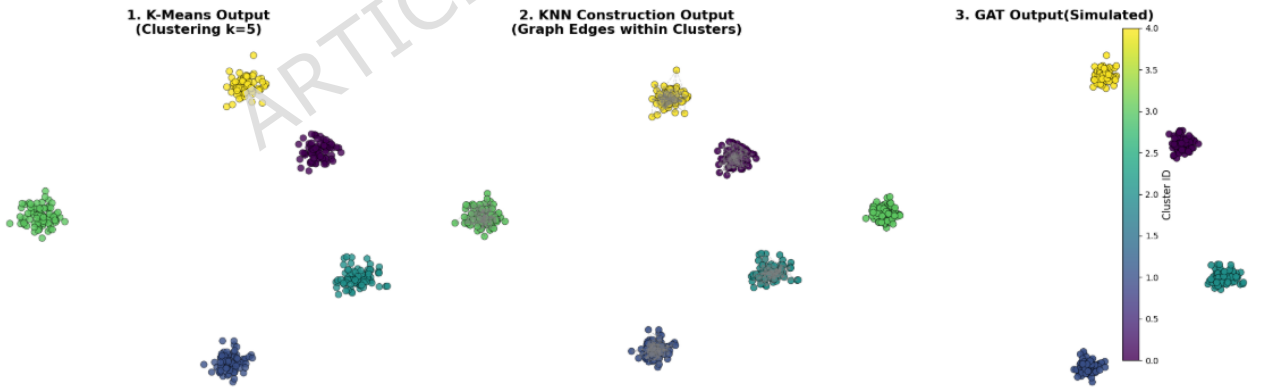


Figure 7. Feature refinement process leading to probabilistic prediction. Panel (a) illustrates the initial partitioning of the raw data using K -means clustering. Panel (b) presents the resulting graph topology, where intra-cluster k -nearest-neighbor (KNN) connections define the local attention neighborhoods. Panel (c) shows the refined embeddings obtained after Graph Attention Network (GAT)-based aggregation, which produces more compact class distributions and enhances the confidence of the subsequent SoftMax-based probability prediction.

The localized graph constructed using the k-nearest neighbors (kNN) approach is further processed by a Graph Attention Network (GAT) [46], as illustrated in Figure 8. The GAT enhances each node's feature representation by selectively aggregating information from its neighbors, effectively learning which neighbors provide the most relevant contextual information. Unlike standard Graph Convolutional Networks (GCNs) [53–56], which rely on fixed edge weights derived from the adjacency matrix, GAT dynamically computes attention coefficients that adaptively modulate the influence of each neighbor.

This adaptive weighting mechanism allows the network to emphasize informative connections and diminish the impact of irrelevant or noisy ones—an essential property in medical image analysis, where subtle features may carry disproportionate diagnostic significance. Compared to other graph-based architectures such as Graph SAGE, GAT offers greater flexibility in modeling heterogeneous interactions among nodes, making it particularly effective for graph-based modeling of retinal image features.

For a given node i , an attention coefficient α_{ij} is computed for each neighboring node $j \in N(i)$, capturing the importance of j 's feature representation relative to i . The attention mechanism is defined by Equation (11):

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [Wh_i \parallel Wh_j]))}{\sum_{k \in N(i)} \exp(\text{LeakyReLU}(a^T [Wh_i \parallel Wh_k]))} \quad (11)$$

Where:

- W is a learnable **weight matrix** applied to transform node features
- a is a learnable **attention vector**
- h_i and h_j are feature vectors of nodes i and j , respectively
- $\parallel$ denotes **concatenation**
- $N(i)$ represents the set of neighbors of node i

To improve the expressiveness and stability of the learned representations, the model employs **multi-head attention**, whereby multiple independent attention mechanisms are executed in parallel. Each head captures a different perspective on neighborhood relationships, and the resulting outputs are typically averaged or concatenated.

Following the attention-based aggregation, the updated embedding h'_i for node i is computed as a weighted sum of the transformed features of its neighbors, followed by a non-linear activation function (Exponential Linear Unit, ELU), as shown in Equation (12):

$$h'_i = \sigma \left(\sum_{j \in N(i)} \alpha_{ij} W h_j \right) \quad (12)$$

Here, σ denotes the ELU activation, which promotes stable learning by allowing negative outputs while avoiding vanishing gradients.

A. Class Prediction via Linear Mapping

Once node representations have been updated through GAT propagation, each node embedding h'_i is mapped into the class space using a fully connected layer parameterized by a learnable weight matrix W_c , followed by a **log-softmax activation** to generate a normalized probability distribution over the C target classes. This is formalized in Equation (13):

$$p_i^{(1)} = \text{softmax}(W_c h'_i) \in [0,1]^C \quad (13)$$

Where h'_i represents the final graph-refined embedding of image i , and $p_i^{(1)}$ denotes the predicted class probability vector from the first (graph-based) branch.

These class probabilities constitute the output of Branch 1 and are later **fused with the predictions** from Branch 2 using an **adaptive stacking ensemble**, enabling the SCGNet framework to combine relational reasoning with spatial representation learning for improved classification accuracy.

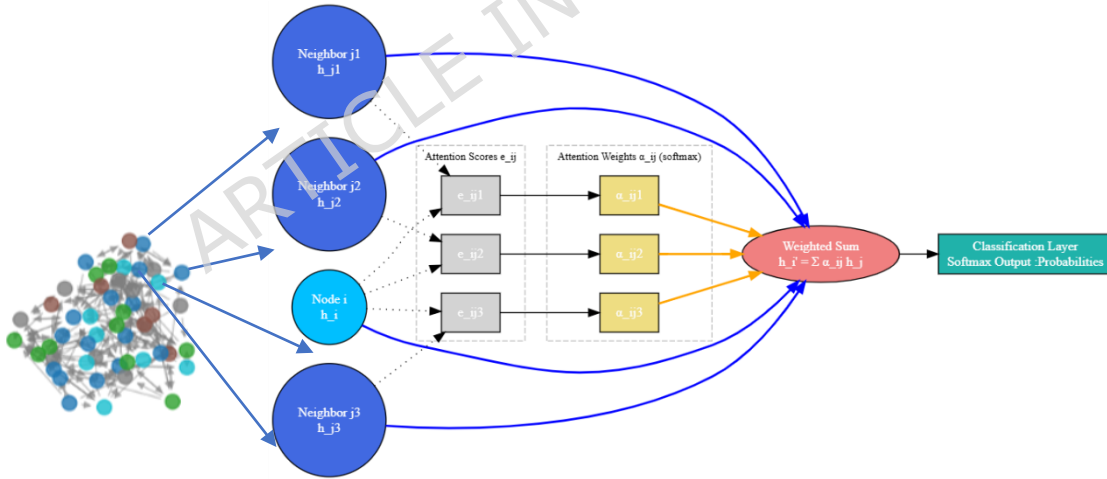


Figure 8: Graph Attention Network Process

2.3.2.Branch 2: ResNet-Based Residual Learning for Hierarchical Feature Representation

The second branch of the SCGNet framework is constructed around a ResNet-50 backbone, a state-of-the-art convolutional neural network architecture renowned for its residual learning capabilities. Introduced to address the vanishing gradient problem in deep neural networks, Residual Networks (ResNet) incorporate identity-based skip connections that allow gradients to propagate directly across layers. This

architectural innovation enables the effective training of deeper models by preserving gradient flow, thereby facilitating stable convergence during optimization. The ability to model hierarchical representations across multiple receptive fields makes ResNet particularly well-suited for medical imaging tasks, where diagnostic accuracy often depends on capturing both fine-grained textures and high-level semantic structures. In this branch, input retinal fundus images are first passed through the **ResNet-50** architecture, which generates rich feature embeddings of dimensionality $D = 2048$. These embeddings encode a comprehensive spectrum of visual patterns—from **low-level features** such as vascular edges and texture variations to **high-level semantic features** such as lesion co-occurrence and structural deformations indicative of diabetic retinopathy (DR).

Given the high dimensionality of the output, Principal Component Analysis (PCA) is applied to project the 2048-dimensional embeddings into a more compact space of $d = 128$ dimensions. This transformation strategically reduces computational overhead while preserving diagnostically informative variance. The use of PCA serves two purposes:

- It filters out redundant or noisy features that may degrade classification performance.
- It significantly improves the computational efficiency of subsequent classification layers, enabling real-time applicability in clinical settings.

The resulting 128-dimensional vectors maintain a high signal-to-noise ratio, capturing the essential pathological indicators necessary for precise DR grading.

The reduced embeddings are then passed through a fully connected output layer, which maps the features into a **probability distribution over the target classes**, corresponding to the five stages of diabetic retinopathy severity. This output constitutes the independent prediction of Branch 2, serving as a complementary signal to the graph-based reasoning conducted in Branch 1.

Several key advantages underline the choice of ResNet for this branch:

- Its ability to **generalize across variable image qualities**, including both low-contrast and artifact-rich inputs
- Proven **transfer learning capabilities**, facilitating robust performance even with moderate-sized labeled datasets
- Compatibility with dimensionality reduction techniques, enabling a balance between feature richness and model efficiency

This design makes Branch 2 particularly valuable in real-world healthcare scenarios, where computational resources may be limited, and rapid, reliable diagnosis is essential. The strategic dimensionality reduction from 2048 to 128 offers a scalable

approach without sacrificing diagnostic depth, reinforcing the suitability of the SCGNet architecture for deployment in diverse clinical environments.

2.3.3. Meta-Classifier Integration: XGBoost for Stacking Ensemble

To synthesize the predictive strengths of the two independent branches—graph-based reasoning (Branch 1) and convolutional feature extraction (Branch 2)—the SCGNet framework employs a stacking ensemble strategy, with XGBoost [57] as the meta-classifier. XGBoost (Extreme Gradient Boosting) is a highly optimized and scalable gradient boosting algorithm known for its strong performance in structured classification tasks. In this context, it is utilized to combine the class probability distributions generated by the two branches into a final, unified prediction.

The choice of XGBoost is motivated by its ability to capture non-linear dependencies between input features and target classes, making it well-suited for learning from heterogeneous information sources. By modeling the complementary contributions of the CNN-based and GNN-based predictions, the meta-classifier effectively enhances the robustness and discriminative capacity of the ensemble. Furthermore, XGBoost incorporates regularization techniques—such as L1 and L2 penalties—that mitigate overfitting and improve generalization to unseen data. This meta-level integration enables the framework to harness both local and relational features, resulting in superior classification performance compared to individual models.

2.3.4. Class Imbalance Handling Through Dynamic Loss Weighting and Adaptive Augmentation

The APTOS and IDRiD datasets exhibit substantial class imbalance, with healthy cases (Grade 0) and mild diabetic retinopathy being far more prevalent than severe proliferative stages (Grade 4). To mitigate this issue without introducing the synthetic distortions often associated with oversampling methods such as SMOTE, this study employs a loss-reweighting strategy in combination with deterministic adaptive augmentation.

During the training of both the Graph Attention Network (GAT) and the ResNet-based classifier, dynamic class weights were incorporated directly into the Negative Log-Likelihood (NLL) loss function. In the implemented framework, class weights, w_c , were computed using the balanced heuristic from `sklearn.utils.class_weight.compute_class_weight`, such that each weight was inversely proportional to the corresponding class frequency in the training data:

$w_c = \frac{N_{\text{total}}}{K \cdot N_c}$ where N_{total} denotes the total number of training samples, K is the number of classes ($K = 5$), and N_c represents the number of samples belonging to class c . This weighting scheme increases the penalty assigned to misclassification of underrepresented severe stages, particularly Grades 3 and 4, thereby encouraging the model to learn more discriminative representations across all severity levels.

In parallel, the augmentation and regularization strategy was designed to improve robustness while preserving subtle pathological details. Instead of relying on random geometric transformations, such as image flipping, which may alter or obscure fine retinal lesions, the proposed framework emphasizes photometric enhancement and controlled noise suppression. Each input image is first processed using the custom Master-Slave Adaptive Notch Filter, which adaptively adjusts the parameters ($\alpha \cdot \sigma$) according to local noise estimates in order to enhance microaneurysms and hemorrhagic patterns consistently across all classes. This preprocessing step is followed by Contrast Limited Adaptive Histogram Equalization (CLAHE) and normalization based on ImageNet statistics. In addition, dropout was introduced to reduce overfitting, with rates of $p = 0.3$ for the GAT module and $p = 0.4$ for the final classifier, while Batch Normalization was applied to further stabilize training. Collectively, this strategy improves sensitivity to rare pathological manifestations while preserving model generalization, as reflected by the macro-averaged F1-score and Cohen's Kappa results.

2.3.5. Experimental Design and Evaluation Metrics

To validate the proposed **SCGNet** framework, a comprehensive experimental protocol was designed, emphasizing both in-domain performance and cross-dataset generalization. The model was initially trained using the APTOS 2019 dataset, which contains 3,662 color fundus images labeled according to the five-stage diabetic retinopathy severity scale. The dataset, as provided on Kaggle, includes predefined patient-level splits for training, validation, and testing, ensuring consistency and reproducibility.

After training, the model was subjected to a multi-tiered evaluation strategy. Internal validation was first conducted using a held-out subset of 366 APTOS images, providing a baseline measure of the model's in-distribution performance. To rigorously assess generalizability, external validation was performed using two demographically and clinically distinct datasets:

- The IDRiD dataset: consisting of 455 high-resolution retinal images collected from an Indian population under controlled imaging conditions.
- A national clinical dataset: comprising 313 fundus images acquired from Moroccan patients in real-world clinical settings using various imaging devices.

This evaluation protocol ensures robust assessment across diverse ethnic backgrounds, imaging conditions, and clinical workflows, thereby validating the model's adaptability and practical utility in global healthcare environments.

The model's performance was measured using both standard classification metrics—such as overall accuracy, macro-averaged precision, and recall—as well as specialized metrics to account for class imbalance. These include:

- Cohen's Kappa: to quantify agreement beyond chance between predicted and actual labels.

- Matthews Correlation Coefficient (MCC): a balanced metric particularly effective for multi-class imbalanced datasets.

All experiments were conducted on NVIDIA Tesla V100 GPUs equipped with 16 GB of VRAM, providing adequate computational resources for efficient training and inference on high-resolution fundus images and comparatively large deep learning architectures. During inference on previously unseen data, graph topology was constructed dynamically and independently for each test batch, thereby ensuring that no test sample was connected to any training node. This design was essential to preserve the validity of the evaluation protocol and to eliminate structural information leakage.

This section describes the computational profiling strategy, the hyperparameter configuration adopted for model training, the precautions implemented to prevent data leakage during graph construction, and the observed efficiency and convergence behavior of the proposed framework.

A. Computational profiling and experimental protocol

At first inspection, the proposed stacking framework, which combines dual deep backbone networks, a graph-based learner, and a meta-ensemble, may appear computationally expensive because of its multi-stage architecture. However, empirical profiling demonstrated that its effective computational burden remained moderate. This efficiency is primarily attributable to the adopted transfer learning strategy, in which the high-capacity pre-trained backbones, namely EfficientNet-B3a and ResNet50, were kept frozen, while only the downstream modules, specifically the proposed GATNet and the ResNet classification head, were optimized.

Computational performance was assessed using three complementary criteria: inference latency, memory utilization, and trainable parameter count. Inference latency was defined as the elapsed time required for a single forward pass. To obtain stable measurements, the system first executed five warm-up iterations, followed by fifty timed iterations. For the Graph Attention Network (GAT), latency was normalized on a per-node basis, whereas for the convolutional backbones it was measured on a per-image basis using a batch size of 16. CUDA synchronization through `torch.cuda.synchronize()` was enforced immediately before and after each timing interval to ensure accurate timing measurements.

GPU memory usage was monitored dynamically using `torch.cuda.memory_allocated()`, and peak active memory was recorded during the backward passes of the GATNet and ResNet head training phases. The number of trainable parameters was computed by summing all parameters for which `requires_grad=True`, thereby explicitly distinguishing optimized components from frozen backbone parameters.

B. Hyperparameter configuration and training dynamics

The training pipeline was designed to achieve a balance between rapid convergence and robust generalization. All trainable components were optimized using the Adam optimizer, with configuration choices adapted to the specific role of each module.

A central hyperparameter in the graph construction stage was the number of clusters used for the K-means topology. Although sensitivity analysis based on Silhouette scores suggested several statistically plausible values in the range $K = 2$ to 12 , the number of clusters was fixed at $K = 5$. This choice was motivated by clinical interpretability, as it directly corresponds to the five diagnostic stages of diabetic retinopathy, from Grade 0 to Grade 4. Within each cluster, graph edges were created by connecting each node to its $k = 5$ nearest neighbors using the Faiss IndexFlatL2 algorithm. This strategy preserved local feature consistency while maintaining a clinically meaningful global organization of the data.

The proposed GATNet consisted of three graph attention layers with a hidden dimensionality of 128 and an attention-head configuration of $[2 \cdot 2 \cdot 1]$. Dropout was applied with a rate of 0.3 in the attention layers and 0.4 at the output stage to mitigate overfitting. The network was trained for a maximum of 300 epochs with a learning rate of 0.01. To address class imbalance, dynamically computed balanced class weights were incorporated directly into the Negative Log-Likelihood (NLL) loss function. Early stopping with a patience of 10 epochs, based on validation loss, was applied to prevent overtraining. Under these settings, convergence was reached at epoch 42. Owing to the graph-structured nature of the data, GATNet training was performed using full-batch gradient descent.

In parallel, a lightweight linear classifier, referred to as the ResNet head, was trained on ResNet50-derived features for 50 epochs using a batch size of 32 and a learning rate of 0.001. Final predictions were obtained through a stacking ensemble in which an XGBoost meta-learner integrated the outputs of the base models. The meta-learner was configured with 100 estimators, a maximum depth of 4, and a learning rate of 0.05.

Before feature extraction and classification, all fundus images were resized to 224×224 pixels. The preprocessing pipeline included the proposed Master-Slave Adaptive Notch Filter with 20 recursive iterations, followed by Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 2.0 and bilateral filtering for additional noise suppression and structural preservation. The resulting features were subsequently standardized and reduced to 128 dimensions using Principal Component Analysis (PCA).

C. Prevention of data leakage in graph construction

To ensure the integrity of the evaluation procedure and to eliminate any possibility of information leakage, graph construction, including clustering and k-nearest neighbor search, was performed independently for the training and testing partitions. This separation was strictly maintained throughout the entire pipeline.

First, all preprocessing transformations requiring parameter estimation, including feature normalization with StandardScaler and dimensionality reduction using PCA, were fitted exclusively on the training set. The learned transformations were then applied to the test set without any refitting. Second, the K-means clustering algorithm used to define graph communities was applied separately to each dataset partition. A dedicated K-means model was fitted on the training features to construct the training graph, while an entirely independent K-means model was fitted on the test features to generate the test graph. No cluster centroids, label assignments, or structural information were transferred between the two partitions.

Finally, the k-nearest neighbor search performed with Faiss was restricted to samples belonging to the same split. Consequently, edges in the training graph linked only training samples, whereas edges in the test graph linked only test samples. During inference, graph topology was therefore generated dynamically and independently for each unseen batch, ensuring that no training node was ever connected to test data. This protocol guaranteed that the model had no access to the structural organization of the test set during training, thereby preserving the reliability and validity of the reported results.

Table 4. *Computational complexity and performance metrics. Inference time is reported per sample (averaged over 50 iterations). Memory usage reflects peak active allocation during the training phase.*

Model Component	Role	Total Parameters	Trainable Parameters	Inference Time (ms)	Peak Training Memory (MB)
EfficientNet-B3a	Feature Extractor	10,696,232	0 (Frozen)	2.37	-
ResNet50	Feature Extractor	23,508,032	0 (Frozen)	2.80	-
GATNet	Graph Classifier	102,159	102,159	0.008	151.99
ResNet Head	Linear Classifier	645	645	< 0.001	153.76
Total Pipeline	Full pipeline	~34.3 M	~0.103 M	< 3.0	~154.0

Table 5. *Summary of Key Hyperparameters and Training Configuration.*

Component	Parameter	Value / Configuration
Clustering	K (Clusters)	5
	Neighbors (k)	5
GATNet	Hidden Dim / Heads	128 / [2,2,1]
	Dropout	0.3 (Attn), 0.4 (Out)
	Learning Rate	0.01 (Adam)
	Early Stopping	Patience = 10
ResNet Head	Epochs / Batch Size	50 / 32
Stacking	Meta-Learner	XGBoost
	Estimators / Depth	100 / 4
Preprocessing	Filter Iterations	20 (Adaptive)
	PCA Components	128

3. Results and Discussion

The proposed SCGNet model demonstrates consistently strong performance across all three datasets—APTOS 2019, IDRiD, and the private Moroccan clinical dataset—highlighting its robustness, cross-domain generalization, and clinical reliability. Detailed classification reports (Tables 6–8) and confusion matrices (Figure 9) provide a comprehensive assessment of its effectiveness in multiclass diabetic retinopathy (DR) classification, spanning the five severity grades (0–4)

3.1. Performance on APTOS 2019 Dataset

The model achieved near-perfect classification on the APTOS dataset, with an overall accuracy of 99% and macro-averaged F1-score of 0.99, indicating exceptional balance between sensitivity and specificity. Notably, precision and recall scores of 1.00 were obtained for several classes, including Mild DR and Proliferative DR, demonstrating that the model is both highly confident and accurate when making positive class predictions.

For instance, the recall of 0.94 for Severe DR implies a small number of false negatives (FN), where the model underestimates disease severity. While this is relatively rare, such errors may delay urgent interventions. In contrast, a precision of 1.00 for the same class indicates that when the model identifies Severe DR, it does so with complete reliability, avoiding unnecessary treatment or referral caused by false positives (FP). These results affirm the model's diagnostic precision under controlled imaging conditions and well-structured class distributions.

3.2. Performance on IDRiD Dataset

The IDRiD dataset presents a more challenging clinical scenario with more variable imaging conditions and subtler early-stage disease indicators. Despite this, the model achieved a commendable accuracy of 95% and a macro-average F1-score of 0.94. However, some degradation is observed in early-stage DR detection. Specifically, the recall of 0.86 for Mild DR suggests that the model occasionally misses cases that exhibit early pathological signs, leading to false negatives that are clinically significant as they limit opportunities for early intervention.

Nevertheless, a precision of 0.95 for Mild DR demonstrates that false alarms are rare. This trade-off illustrates a conservative decision boundary, where the model prioritizes specificity over sensitivity in early grades—an acceptable clinical stance when the cost of false positives (e.g., unnecessary anxiety or referrals) must be minimized. These observations underscore the sensitivity-specificity trade-off inherent in multiclass medical classification models and highlight areas for improvement, such as threshold calibration or data augmentation for early-stage classes.

3.3. Performance on Private Clinical Dataset

On the private Moroccan dataset, which reflects real-world variability in image quality and acquisition protocols, the model maintained strong generalization, achieving an overall accuracy of 96% and macro-averaged F1-score of 0.96. Particularly noteworthy is the model's performance on the Proliferative DR class, where it achieved precision of 1.00—indicating no false positives, which is critical in avoiding overdiagnosis of severe disease. However, the recall of 0.91 for this class reveals the presence of false negatives—i.e., a few true proliferative cases incorrectly predicted as Severe DR. While both classes indicate advanced disease, the clinical management and urgency may differ, making this a relevant error class for further refinement.

Conversely, Severe DR and Normal classes both showed near-perfect recall (0.99), indicating that the model seldom overlooks either healthy cases or advanced disease—a desirable property in clinical screening contexts where minimizing diagnostic risk is essential.

3.4. Cross-Dataset Insights and Model Behavior

Across all three datasets, high F1-scores (≥ 0.94 macro-average) confirm that SCGNet maintains a strong balance between precision and recall, demonstrating no class bias or domain-specific overfitting. Misclassifications, as revealed by the confusion matrices (Figure 9), typically occur between adjacent DR grades, such as Mild vs. Moderate DR. This pattern is consistent with clinical reality, where even expert ophthalmologists face challenges in differentiating between subtle textural variations across consecutive severity levels.

These results also validate the design rationale of SCGNet: the fusion of CNN-based hierarchical feature extraction and GAT-based relational reasoning preserves diagnostically critical features while suppressing inter-class confusion caused by imaging variability or lesion ambiguity. The model's recursive preprocessing pipeline further contributes to this robustness by enhancing subtle lesion visibility and reducing image noise.

Furthermore, the use of Cohen's Kappa and Matthews Correlation Coefficient (MCC) as complementary metrics ensures a statistically rigorous evaluation, especially in the presence of class imbalance. High Kappa values reflect strong agreement between predictions and ground truth beyond chance, while high MCC confirms that the model performs well across all classes, not just the majority ones.

While the 99.45% accuracy achieved on APTOS is remarkably high, it is consistent with the design of the proposed framework and should be interpreted in the context of its methodological safeguards. The result is likely driven by the synergy between the robust preprocessing strategy and the strict patient-level partitioning protocol, the latter of which effectively prevents overfitting caused by subject overlap between training and testing subsets. Moreover, the preservation of strong performance on the external IDRiD dataset and the private dataset further indicates that the model does not merely exploit dataset-specific characteristics, but rather learns clinically meaningful and transferable representations. This cross-dataset consistency strengthens the credibility of the reported result and highlights the generalization capacity of the proposed method.

Table 6: Classification report for each DR class on the test set for APTOS dataset

Classes	Precision	recall	F1-score	support
Normal	1.00	0.99	1.00	199
Mild DR	1.00	1.00	1.00	30
Moderate DR	0.98	1.00	0.99	87
Severe DR	1.00	0.94	0.97	17
Proliferative DR	1.00	1.00	1.00	33
Accuracy			0.99	366
Macro avg	1.00	0.99	0.99	366

Weighted avg	0.99	0.99	0.99	366
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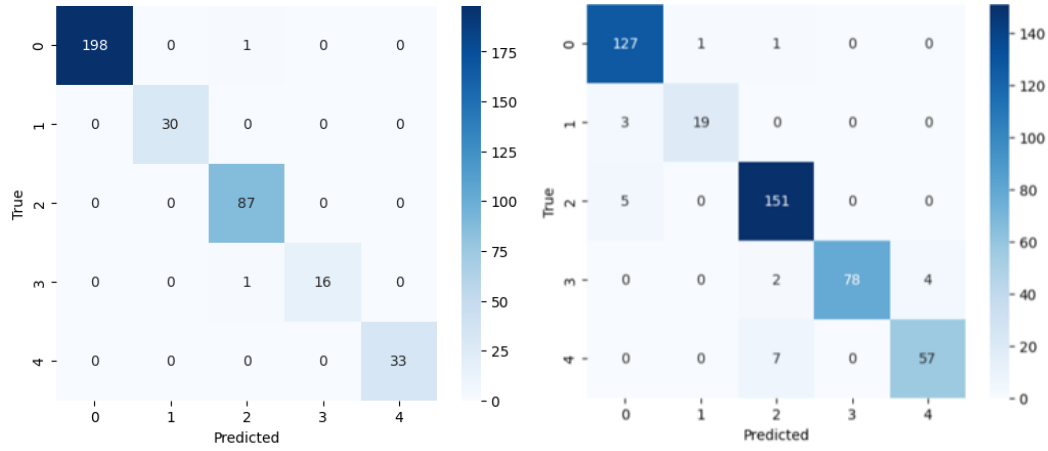
Table 7: Classification report for each DR class on the test set for IDRID dataset

Classes	Precision	recall	F1-score	Support
Normal	0.94	0.98	0.96	129
Mild DR	0.95	0.86	0.90	22
Moderate DR	0.94	0.97	0.95	156
Severe DR	1.00	0.93	0.96	84
Proliferative DR	0.93	0.89	0.91	64
Accuracy			0.95	455
Macro avg	0.95	0.93	0.94	455
Weighted avg	0.95	0.95	0.95	455

Table 8: Classification report for each DR class on the test set for the Private dataset.

Classes	Precision	recall	F1-score	Support
Normal	0.98	0.99	0.98	83
Mild DR	0.97	0.90	0.94	41
Moderate DR	0.91	0.94	0.92	84
Severe DR	0.98	0.99	0.98	83
Proliferative DR	1.00	0.91	0.95	22
Accuracy			0.96	313
Macro avg	0.97	0.95	0.96	313
Weighted avg	0.96	0.96	0.96	313

Aptos Dataset(a) IDRID Dataset(b)



Private Dataset(c)

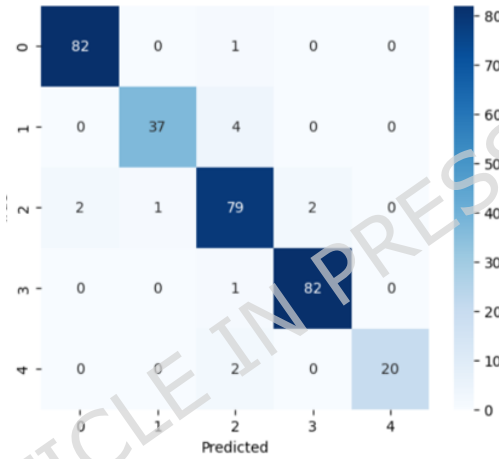


Figure.9. Confusion matrix of the DR classification model on the independent test set (N = 366 images) (a), set (N = 455 images) IDRID Dataset(b) and (N = 313 images) National Private Dataset(c) showing per-class classification performance with counts and percentages.

3.5. Comparative Performance Analysis with Baseline Models

To validate the architectural design of SCGNet, an ablation study was conducted to examine the contribution of four principal components: the preprocessing pipeline, the graph-based relational reasoning module (GAT branch), the ResNet branch, and the stacking ensemble. Table 9 reports the results of this analysis and reveals the extent of performance degradation associated with the removal of each component. In particular, omitting the stacking ensemble resulted in a substantial decline in accuracy, thereby confirming the benefit of integrating the two complementary branches within a unified framework. This trend is further illustrated in Figure 10, where the complete SCGNet architecture (green bar) achieves superior performance compared with all ablated variants, especially with respect to Macro F1-score.

Table 9. Results of the ablation analysis evaluating the contribution of four principal components of the proposed SCGNet framework: the preprocessing pipeline, the graph-based relational reasoning module (GAT branch), the ResNet branch, and the stacking ensemble.

Model Variant	Accuracy (%)	Macro F1	Kappa
SCGNet (Full)	99.45	0.99	0.99
Without Stacking (Only ResNet)	78.69	0.55	0.66
Without Stacking (Only GAT)	51.64	0.32	0.30
Without Clustering (Random Graph)	85.0	0.70	0.75
WithoutPreprocessing (Raw Images)	81.0	0.60	0.80

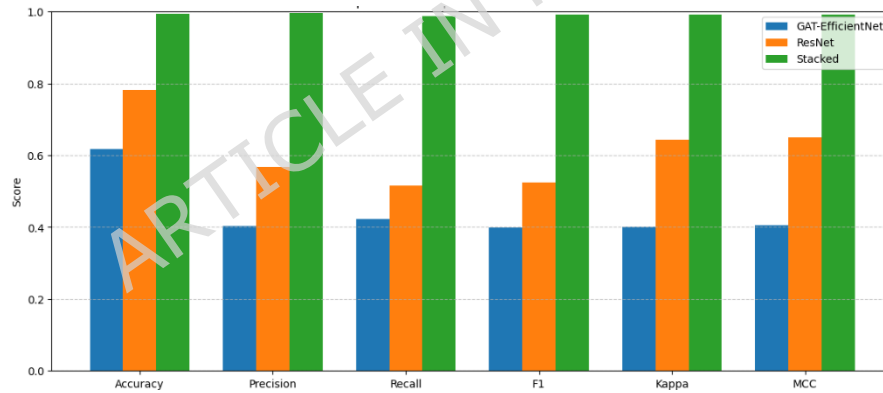


Figure.10. Comparative Performance Graph Analysis with Baseline Models

Furthermore, the final proposed model was compared with state-of-the-art methods to position its performance within the current literature. Table.10. presents a comparative evaluation of representative deep learning models for diabetic retinopathy (DR) classification on the APTOS 2019 dataset. The analysis includes widely adopted architectures such as EfficientNet, ResNet, InceptionV3, and DenseNet, alongside the proposed SCGNet model. For each method, key performance metrics are reported: overall accuracy, macro-averaged precision, F1-score, and Cohen's Kappa coefficient, allowing for a multidimensional assessment of classification performance across all five DR severity levels.

As illustrated in the table, the SCGNet model outperforms all baseline approaches across the majority of metrics. Specifically, it achieves the highest accuracy and F1-score, indicating superior capability in correctly identifying both early and advanced DR stages. These improvements are particularly meaningful in a multiclass setting, where preserving sensitivity to mild disease while maintaining precision for severe cases is a complex challenge.

Furthermore, the model's elevated Kappa score reflects strong agreement with expert-labeled ground truth, reinforcing its diagnostic reliability. The combination of convolutional and graph-based components—complemented by a stacking ensemble—appears to significantly enhance the model's discriminative power, particularly in capturing subtle inter-class differences that traditional CNNs may overlook.

Overall, this comparative analysis substantiates the efficacy and generalization of the proposed model, establishing it as a robust and competitive solution for real-world DR screening applications.

Table 10. Comparison of accuracy and F1-score of recent DR classification models on the APTOS 2019 dataset

Study	Model / Method	Dataset	No. of classes	Accuracy (%)	Precision (%)	F1-score (%)	kappa score (%)
Yi et al. 2020[58]	EfficientNet-B3 + Ensemble Learning	APTOS	- 5	93.55	-	-	91.93
Sharmin Majumder 2021 [59]	MXception + Data Augmentation	APTOS EyePACS	5	81.00 82.00	70.00 69.00	64.00 66.00	81.00 88.00
Kobat et al..2022 [60]	DenseNet201	- APTOS	5	87.43%	-	-	-
Sajid et al. 2023[61]	DenseNet121	Combined Dataset	5	96.00	93.70	94.50	94.90
Alwakid et al. 2023 [62]	CLAHE + ESRGAN + EfficientNetB5	APTOS	5	79.23	78.00	79.00	
Proposed Model	SCGNet	-APTOS -IDRID - PRIVATE DATASET	5	99.00 95.00 96.00	100 95.00 97.00	99.12 94.00 96.00	99.13 93.19 94.55

3.6. Statistical Robustness and Confidence Interval Analysis

To assess the robustness of the proposed stacked model (GAT + ResNet + XGBoost) beyond a single performance estimate, a statistical reliability analysis was performed using a 95% confidence interval (CI) for classification accuracy. While the observed accuracy offers a direct measure of predictive performance, it does not capture the uncertainty inherent in evaluating the model on a finite test sample. Therefore, confidence interval estimation was used to provide a more rigorous assessment of the stability and generalizability of the reported results.

On the test set comprising $N = 366$ samples, the model achieved an observed accuracy of 0.9590. Using the normal approximation of the binomial proportion with a Z-value of 1.96, the standard error was estimated at 0.01036, corresponding to a margin of error of ± 0.0203 . This yielded a 95% confidence interval of $[0.9387, 0.9793]$ for the true accuracy.

The obtained interval suggests that the predictive performance of the proposed framework is expected to remain above 93.87% under similar evaluation conditions, with 95% confidence. The narrow confidence bounds indicate limited statistical variability and reinforce the conclusion that the model performance is stable rather than incidental to a particular data partition. Overall, this analysis provides additional evidence of the robustness and generalization capacity of the proposed method.

4. Conclusion and Future Directions

This study presents SCGNet, a novel ensemble-based architecture that integrates Graph Attention Networks (GAT) and ResNet within a unified framework for diabetic retinopathy (DR) classification from retinal fundus images. The model is further strengthened by a robust, multi-stage preprocessing pipeline that enhances image quality and preserves diagnostically relevant features. By combining convolutional visual feature extraction with inter-sample relational modeling, SCGNet effectively leverages both local image-level information and contextual dependencies across samples—an approach that has demonstrated considerable benefits for diagnostic accuracy.

Trained on the APTOS 2019 dataset and evaluated across three diverse datasets—APTOS, IDRiD, and a real-world private clinical dataset—SCGNet achieved near-perfect accuracy on the source domain and consistently high precision, recall, and inter-rater agreement metrics during external validation. These results underscore the model's ability to generalize across datasets with heterogeneous imaging conditions, resolutions, and annotation protocols—a critical capability for real-world clinical deployment.

A key factor in the model's robustness lies in its use of graph attention mechanisms, which enable adaptive aggregation of information from the most informative neighbors. This facilitates the capture of richer contextual cues, allowing SCGNet to distinguish between fine-grained stages of DR—even under domain shifts—where traditional convolutional models may falter. The architectural design also lends itself well to broader applications beyond DR, offering a scalable framework for other medical imaging tasks that could benefit from relational reasoning, such as tumor detection or organ segmentation.

From a clinical deployment perspective, SCGNet Model was explicitly designed to achieve high computational efficiency. With an inference latency of less than 3.0 ms per image and only approximately 0.1 million trainable parameters, the model is sufficiently lightweight to operate on standard clinical workstations, without the need for high-performance GPU clusters. This efficiency is particularly advantageous in screening settings, where scalability, processing speed, and hardware availability are critical considerations.

Beyond its computational advantages, the dual-branch design of SCGNet also provides an interpretable framework for decision support. The CNN branch is responsible for capturing salient spatial and morphological features from the retinal image, while the GAT branch exploits graph-based attention mechanisms to model relationships among similar samples in the learned feature space. As a result, the latter may offer a context-aware form of reasoning by identifying neighboring cases with related characteristics, thereby complementing the image-based prediction with relational evidence that may be clinically informative.

Despite these promising characteristics, translation into routine practice will require further validation beyond retrospective experiments. Future regulatory approval processes, including FDA clearance or CE marking, will depend on prospective clinical trials designed to assess performance under real-time screening conditions. Such studies will be essential to verify the robustness, safety, and clinical utility of the proposed framework in practical deployment scenarios.

Future Work

Future efforts will focus on translating SCGNet into operational clinical environments, assessing its performance on larger and more diverse datasets collected under varying real-world conditions. This includes evaluating its adaptability across different imaging devices, healthcare settings, and population demographics. Integration into routine diagnostic workflows will enable assessment of not only classification performance but also computational efficiency, interpretability, and clinical usability.

Planned ablation studies will further dissect the individual contributions of each architectural component—particularly the dual-branch design and stacking ensemble strategy—to identify the most influential mechanisms driving performance. These insights will guide targeted architectural refinements aimed at

improving reliability, reducing inference latency, and optimizing model deployment on resource-constrained systems.

Additionally, we envision extending the SCGNet framework into a multimodal diagnostic system by incorporating complementary patient data—such as age, diabetes duration, HbA1c levels, comorbidities, and DR history—alongside fundus images. This integration of clinical and visual data would support personalized risk stratification, enhance predictive accuracy, and improve early detection and longitudinal monitoring of diabetic retinopathy progression. Ultimately, such a system could serve as a foundational platform for comprehensive, AI-assisted ophthalmic care, capable of functioning across diverse healthcare infrastructures.

Authors' contributions

All authors read and approved the final manuscript.

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Availability of data and materials

Kaggle public version of Aptos dataset: <https://www.kaggle.com/datasets/mariaherrerot/aptos2019>[63]
Kaggle public Dataset IDRID: <https://www.kaggle.com/datasets/mariaherrerot/idrid-dataset> [64] Private Dataset: available upon request from the authors.[32]

Declarations

Ethical approval

This study was approved by the Local Ethics Committee of Sheikh Zaid Hospital, Rabat, Morocco, under Project No. CEFCZ/PR/03/06/24. The study was supported by the Sheikh Zaid Foundation and was conducted in accordance with relevant ethical guidelines for research involving clinical data.

Consent to participate

The requirement for informed consent was waived by the Local Ethics Committee of Sheikh Zaid Hospital, Rabat, Morocco, due to the retrospective and anonymized nature of the clinical data used in this study.

Consent to publish

The manuscript does not contain any personally identifiable information or individual participant data. Therefore, consent for publication is not applicable.

Competing interests

The authors declare that they have no competing interests.

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References

- [1] World Health Organization, WHO Official Website, 2023. [Online]. Available: <https://www.who.int/>.
- [2] International Diabetes Federation, IDF Diabetes Atlas, 10th ed., Brussels, Belgium, 2021. <https://diabetesatlas.org>.
- [3] E. Kohner, V. Patel, and S. M. Rassam, "Role of blood flow and impaired autoregulation in the pathogenesis of diabetic retinopathy," *Diabetes*, vol. 44, no. 6, pp. 603-608, 1995, doi: 10.2337/diab.44.6.603.
- [4] X. Li and X. Li, "The antidepressant effect of light therapy from retinal projections," *Neuroscience Bulletin*, vol. 34, no. 2, pp. 359-368, 2018, doi: 10.1007/s12264-018-0210-1.
- [5] M. Atkinson and N. Maclaren, "What causes diabetes?," *Scientific American*, vol. 263, pp. 62-63, 1990, doi: 10.1038/scientificamerican0790-62.
- [6] M. Nentwich and M. Ulbig, "Diabetic retinopathy - ocular complications of diabetes mellitus," *World Journal of Diabetes*, vol. 6, no. 3, pp. 489-499, 2015, doi: 10.4239/wjd.v6.i3.489.
- [7] Deepa V., Kumar C.S., Cherian T. Ensemble of multi-stage deep convolutional neural networks for automated grading of diabetic retinopathy using image patches. *Journal of King Saud University - Computer and Information Sciences*. 2022;34(8):6255-6265. <https://doi.org/10.1016/j.jksuci.2021.05.009>
- [8] H. Liu, K. Yue, S. Cheng, C. Pan, J. Sun, and W. Li, "Hybrid model structure for diabetic retinopathy classification," *Journal of Healthcare Engineering*, vol. 2020, pp. 1-12, 2020, doi: 10.1155/2020/8840174.
- [9] Sugeno A., Ishikawa Y., Ohshima T., Muramatsu R. Simple methods for the lesion detection and severity grading of diabetic retinopathy by image processing and transfer learning. *Computers in Biology and Medicine*. 2021;137:104804. DOI: 10.1016/j.combiomed.2021.104795
- [10] P. Lanzetta, V. Sarao, P. H. Scanlon, J. Barratt, M. Porta, F. Bandello, and A. Loewenstein, "Fundamental principles of an effective diabetic retinopathy screening program," *Acta Diabetol.*, vol. 57, no. 7, pp. 785-798, Mar. 2020, doi: 10.1007/s00592-020-01506-8.
- [11] Suedumrong C., Leksakul K., Wattana P., Chaopaisarn P. Application of deep convolutional neural networks vgg-16 and googlenet for level diabetic retinopathy detection. In: *Lecture Notes in Networks and Systems*. 2022;359:56-65. https://link.springer.com/chapter/10.1007/978-3-030-89880-9_5

- [12] W. Almattar, H. Lugman, and F. A. Khan, "Diabetic retinopathy grading review: Current techniques and future directions," *Image and Vision Computing*, vol. 139, 104821, 2023, doi: 10.1016/j.imavis.2023.104821.
- [13] D. S. W. Ting, G. C. M. Cheung, and T. Y. Wong, "Diabetic retinopathy: global prevalence, major risk factors, screening practices and public health challenges: a review," *Clinical & Experimental Ophthalmology*, vol. 44, no. 4, pp. 260-277, 2016, doi: 10.1111/ceo.12696.
- [14] V. Gulshan, R. P. Rajan, K. Widner, et al., "Performance of a deep-learning algorithm vs manual grading for detecting diabetic retinopathy in India," *JAMA Ophthalmology*, vol. 137, no. 9, pp. 987-993, 2019, doi: 10.1001/jamaophthalmol.2019.2004.
- [15] N. Asiri, M. Hussain, F. Adel, and N. Alzaidi, "Deep learning based computer-aided diagnosis systems for diabetic retinopathy: A survey," *Artificial Intelligence in Medicine*, vol. 99, 101701, 2019, doi: 10.1016/j.artmed.2019.07.009.
- [16] V. Gulshan, et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402-2410, 2016, doi: 10.1001/jama.2016.17216.
- [17] M. D. Abramoff, et al., "Improved automated detection of diabetic retinopathy on a publicly available dataset through integration of deep learning," *Investigative Ophthalmology & Visual Science*, vol. 57, no. 13, pp. 5200-5206, 2016, doi: 10.1167/iovs.16-19964.
- [18] H. Pratt, et al., "Convolutional neural networks for diabetic retinopathy," *Procedia Computer Science*, vol. 90, pp. 200-205, 2016, doi: 10.1016/j.procs.2016.07.014.
- [19] D. S. W. Ting, et al., "Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes," *JAMA*, vol. 318, no. 22, pp. 2211-2223, 2017, doi: 10.1001/jama.2017.18152.
- [20] M. Ikram and A. Imran, "ResViTFusionNetModel: An explainable hybrid architecture for diabetic retinopathy classification," *Medical Image Analysis*, vol. 92, 103065, 2024, doi: 10.1016/j.compbimed.2025.109656.
- [21] K. Fatema, I. U. Khan, S. Azam, et al., "A lightweight robust deep learning model gained high accuracy in classifying diabetic retinopathy images," *IEEE Access*, 2023, doi: 10.1109/ACCESS.2023.3272228.
- [22] O. Mecili, H. Barkat, F. Nouioua, et al., "An efficient explainable deep neural network classifier for diabetic retinopathy detection," *International Journal of Computers and Applications*, 2024, doi: 10.1080/1206212X.2024.2389342.
- [23] D. Bouslimi, Y. Bouslimi, A. K. Echi, et al., "Deep learning for diabetic retinopathy early detection and severity grading," *IEEE ATSIP*, 2024, doi: 10.1109/ATSIP62566.2024.10638995.
- [24] Y. Guerbai, W. Bouaraki, A. Saibi, et al., "Deep learning techniques for diabetic retinopathy classification: A focus on VGG16 and EfficientNetB0," *South Florida Journal of Development*, vol. 5, no. 10, 2024, doi: 10.46932/sfjdv5n10-027.

- [25] I. K. Gupta, S. Patil, S. Mahadevkar, K. Kotecha, A. K. Mishra, and J. J. P. C. Rodrigues, "Retinal fundus imaging-based diabetic retinopathy classification using transfer learning and fennec fox optimization," *MethodsX*, vol. 14, 103232, 2025, doi: 10.1016/j.mex.2025.103232.
- [26] D. S. Renu and K. S. Saji, "Hybrid deep learning framework for diabetic retinopathy classification with optimized attention AlexNet," *Computers in Biology and Medicine*, vol. 190, 110054, 2025, doi: 10.1016/j.combiomed.2025.110054.
- [27] C. A. Girkin et al., "Variation in optic nerve and macular structure with age and race with spectral-domain optical coherence tomography," *Ophthalmology*, vol. 118, no. 12, pp. 2403–2408, Dec. 2011. doi: 10.1016/j.ophtha.2011.06.013.
- [28] A. Gichoya, et al., "AI recognition of patient race in medical imaging: a modelling study," *The Lancet Digital Health*, 2022 Jun;4(6):e406-e414, DOI: 10.1016/S2589-7500(22)00063-2.
- [29] Y. Lei, S. Lin, Z. Li, Y. Zhang, and T. Lai, "GNN-fused CapsNet with multi-head prediction for diabetic retinopathy grading," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 107994, 2024. DOI: 10.1016/j.engappai.2024.107994
- [30] V. K. Pandugula, A. Choudhary, R. Uyyala, and P. Vurubindi, "Hybrid CNN-Graph Attention Networks for Diabetic Retinopathy Grading: A Multimodal Feature Fusion Approach," in *2025 3rd International Conference on Inventive Computing and Informatics (ICICI)*, Coimbatore, India, 2025, pp. 953–958. DOI: 10.1109/ICICI65870.2025.11069687
- [31] K. V. Deepsahith, S. Basineni, K. Bangipavan, A. Sherly, S. Brindha, and G. Subramanian, "Graph-enhanced multimodal fusion of vascular biomarkers and deep features for diabetic retinopathy detection," *Frontiers in Artificial Intelligence*, vol. 8, 2026. DOI: 10.3389/frai.2025.1731633
- [32] Bahia OUZZANI CHAHDI Faculty of medicine and pharmacy Mouhammed V University, Rabat, Morocco mail:bahia.ouazzani_chahdi@um5.ac.ma
- [33] Y. Ge, L. Yang, and X. Ma, "A harmonic compensation method for SPMSM sensorless control based on the orthogonal master-slave adaptive notch filter," *IEEE Trans. Power Electron.*, vol. 36, no. 10, pp. 11701–11711, Oct. 2021.
- [34] J. Astola and P. Kuosmanen, *Fundamentals of Nonlinear Digital Filtering*, Boca Raton, FL, USA: CRC Press, 1997.
- [35] A. H. Sayed, *Fundamentals of Adaptive Filtering*, Hoboken, NJ, USA: Wiley-Interscience, 2003.
- [36] P. Bankhead, C. N. Scholfield, J. G. McGeown, and T. M. Curtis, "Fast retinal vessel detection and measurement using wavelets and edge location refinement," *PLoS One*, vol. 7, no. 3, p. e32435, Mar. 2012, doi: 10.1371/journal.pone.0032435.
- [37] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA, USA: Academic Press, 1994, pp. 474–485. <https://doi.org/10.1016/B978-0-12-336156-1.50061-6>.
- [38] R. Pires, H. F. Jelinek, and J. Wainer, "Color retinal image enhancement using CLAHE," in *Proc. 2013 International Conference on ICT for Smart Society (ICISS)*, Jun. 2013, pp. 151–155, doi: 10.1109/ICTSS.2013.6588092.

- [39] C. Tomasi and R. Manduchi, "Bilateral filtering for gray and color images," in Proc. IEEE Int. Conf. Comput. Vis. (ICCV), 1998, pp. 839-846, doi: 10.1109/ICCV.1998.710815.
- [40] S. Paris, P. Kornprobst, J. Tumblin, and F. Durand, "Bilateral filtering: Theory and applications," *Foundations and Trends in Computer Graphics and Vision*, vol. 4, no. 1, pp. 1-73, 2009, DOI: 10.1561/06000000020.
- [41] F. Durand and J. Dorsey, "Fast bilateral filtering for the display of high-dynamic-range images," *ACM Transactions on Graphics (TOG)*, vol. 21, no. 3, pp. 257-266, 2002, doi: 10.1145/566654.566574.
- [42] A. Krizhevsky, I. Sutskever, and G. Hinton, "ImageNet classification with deep convolutional neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2012, pp. 1097-1105.
- [43] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 770-778, doi: 10.1109/CVPR.2016.90.
- [44] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in Proc. Int. Conf. Learn. Represent. (ICLR), 2017. [Online]. Available: <https://arxiv.org/abs/1609.02907>.
- [45] P. Veličković, et al., "Graph attention networks," in Proc. Int. Conf. Learn. Represent. (ICLR), 2018. [Online]. Available: <https://arxiv.org/abs/1710.10903>.
- [46] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. 36th Int. Conf. Machine Learning (ICML), 2019. <https://proceedings.mlr.press/v97/tan19a.html>.
- [47] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed., New York, NY, USA: Springer, 2002.
- [48] K. Pearson, "On lines and planes of closest fit to systems of points in space," *Philosophical Magazine*, vol. 2, no. 11, pp. 559-572, 1901, doi: 10.1080/14786440109462720.
- [49] H. Abdi and L. J. Williams, "Principal component analysis," *Wiley Interdisciplinary Reviews: Computational Statistics*, vol. 2, no. 4, pp. 433-459, 2010, doi: 10.1002/wics.101.
- [50] D. Arthur and S. Vassilvitskii, "k-means++: The advantages of careful seeding," in Proc. ACM-SIAM Symp. Discrete Algorithms (SODA), 2007, pp. 1027-1035.
- [51] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Transactions on Information Theory*, vol. 13, no. 1, pp. 21-27, 1967, doi: 10.1109/TIT.1967.1053964.
- [52] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535-547, 2021, doi: 10.1109/TBDATA.2019.2921572.
- [53] S. Zhang, H. Tong, J. Xu, and R. Maciejewski, "Graph convolutional networks: a comprehensive review," *Comput. Soc. Netw.*, vol. 6, no. 1, Art. 11, 2019. doi: 10.1186/s40649-019-0069-y.
<https://computationalsocialnetworks.springeropen.com/articles/10.1186/s40649-019-0069-y>

- [54] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 1, pp. 4–24, 2021, doi: 10.1109/TNNLS.2020.2978386.
- [55] M. M. Bronstein, J. Bruna, Y. LeCun, A. Szlam, and P. Vandergheynst, "Geometric deep learning: Going beyond Euclidean data," *IEEE Signal Processing Magazine*, vol. 34, no. 4, pp. 18–42, 2017, doi: 10.1109/MSP.2017.2693418.
- [56] Z. Zhang, P. Cui, and W. Zhu, "Deep learning on graphs: A survey," *IEEE Transactions on Knowledge and Data Engineering*, vol. 34, no. 1, pp. 249–270, 2022, doi: 10.1109/TKDE.2020.2981333.
- [57] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [58] S.-L. Yi, X.-L. Yang, T.-W. Wang, F.-R. Sun, X. Xiong, and J.-F. Huang, "Diabetic Retinopathy Diagnosis Based on RA-EfficientNet," *Appl. Sci.*, vol. 11, no. 22, 2021, doi:10.3390/app112211035.
- [59] S. Majumder and N. Kehtarnavaz, "Multitasking Deep Learning Model for Detection of Five Stages of Diabetic Retinopathy," *arXiv:2103.04207*, Mar. 2021. <https://arxiv.org/abs/2103.04207>.
- [60] S. G. Kobat, N. Baygin, E. Yusufoglu, M. Baygin et al., "Automated Diabetic Retinopathy Detection Using Horizontal and Vertical Patch Division-Based Pre-Trained DenseNET with Digital Fundus Images," *Diagnostics*, vol. 12, no. 8, p. 1975, 2022. doi:10.3390/diagnostics12081975.
- [61] M. Z. Sajid, M. F. Hamid, A. Youssef et al., "DR-NASNet: Automated System to Detect and Classify Diabetic Retinopathy Severity Using Improved Pretrained NASNet Model," *Diagnostics*, vol. 13, no. 16, p. 2645, 2023. doi:10.3390/diagnostics13162645.
- [62] G. Alwakid, W. Gouda, and M. Humayun, "Enhancement of Diabetic Retinopathy Prognostication Using Deep Learning, CLAHE, and ESRGAN," *Diagnostics*, vol. 13, no. 14, p. 2375, 2023. doi:10.3390/diagnostics13142375. (Also: *Digital Health*, 2023).
- [63] Kaggle public version of Aptos dataset: <https://www.kaggle.com/datasets/mariaherrerot/aptos2019>.
- [64] Kaggle-public DATASET IDRID: <https://www.kaggle.com/datasets/mariaherrerot/idrid-dataset>.